



Stress Test Pipeline — Test Results

Automated Test Documentation for Model Governance

MKM Research Labs

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1 Test Results — Stress Test Pipeline (MKM-ST-001)

Tests run on **29 March 2026 at 10:29:14**.

Table 1: Test Results Summary — Stress Test Pipeline (MKM-ST-001)

Total Tests	107
Passed	107
Failed	0
Skipped	0
Duration	16.62s

Table 2: Detailed Test Results — Stress Test Pipeline

Test Description	Acceptance Criteria	Result	Time
All already trained	<code>result["trained"] == 0;</code> <code>result["skipped"] == 3</code>	Pass	0.003s
After the first gauge trains, ETA should appear in output.	<code>"ETA" in captured</code>	Pass	0.234s
Missing gauge json	—	Pass	0.000s
Missing sequences	—	Pass	0.002s
No valid gauges	<code>result["trained"] == 0</code>	Pass	0.001s
One gauge already trained, one succeeds, one fails.	<code>result["trained"] == 1;</code> <code>result["failed"] == 1 (+1 more)</code>	Pass	0.157s
Successful training	<code>result["trained"] == 3;</code> <code>result["failed"] == 0 (+2 more)</code>	Pass	0.236s
Training failure is caught	<code>result["trained"] == 0;</code> <code>result["failed"] == 3 (+1 more)</code>	Pass	0.290s
Counts default to zero	<code>s["alert_sequences"] == 0;</code> <code>s["warning_sequences"] == 0</code> (+1 more)	Pass	0.000s
Num gauges	<code>s["num_gauges"] == 4</code>	Pass	0.000s
Num sequences	<code>s["num_sequences"] == 7</code>	Pass	0.001s
Returns required keys	<code>key in s</code>	Pass	0.001s
When one gauge's classifier fails, pipeline continues.	<code>"classifier failed" in out;</code> <code>result["num_gauges"] == 2</code>	Pass	0.038s
Dict format	<code>len(result) == 2</code>	Pass	0.000s
Empty list	<code>._extract_gauges({"flood_gauges": []}) == []</code>	Pass	0.000s
List format	<code>result == [_G1, _G2]</code>	Pass	0.000s
Missing key returns empty	<code>._extract_gauges({}) == []</code>	Pass	0.000s

Test Description	Acceptance Criteria	Result	Time
When gaugehd/ has baseline data, the 'Loaded' message appears.	"gauge baselines from gaugehd" in out or "heuristic base ...	Pass	0.039s
Gauge summary dir written	sg_dir.is_dir()	Pass	0.000s
Gauge summary gauge ids	len(d["gauge_ids"]) == 3	Pass	0.000s
Gauge summary num gauges	d["num-gauges"] == 3	Pass	0.000s
Gauge summary schema version	meta.exists(); d["schema_version"] == SCHEMA_VERSION_SPATIAL	Pass	0.002s
Gauge summary sequence count	len(gauge_files) == 3	Pass	0.000s
Sequences file loadable	len(seqs) == 40	Pass	0.001s
Sequences json written	(gauge_dir / SEQUENCES_FILENAME).exists()	Pass	0.000s
Summary json written	(gauge_dir / SUMMARY_FILENAME).exists()	Pass	0.000s
Alert count non negative	full_run["alert_sequences"] >= 0	Pass	0.000s
Elapsed positive	full_run["elapsed_seconds"] > 0	Pass	0.000s
Num gauges	full_run["num-gauges"] == 3	Pass	0.000s
Num sequences	full_run["num-sequences"] == 40	Pass	0.000s
Returns dict	isinstance(full_run, dict)	Pass	0.000s
Severe lte warning	full_run["severe_sequences"] <= full_run["warning_sequenc...	Pass	0.000s
Type counts present	"type_counts" in full_run; sum(full_run["type_counts"].values()) == 40	Pass	0.000s
Warning lte alert	full_run["warning_sequences"] <= full_run["alert_sequences"]	Pass	0.000s
Batch generator populates category	all(s.intensity_category in valid for s in seqs)	Pass	0.001s
From dict missing category defaults empty	reconstructed.intensity_category == ""	Pass	0.000s
From dict round trip	reconstructed.intensity_category == s.intensity_category	Pass	0.000s
Serialized sequences have category	all(s.intensity_category in valid for s in seqs)	Pass	0.001s
To dict includes intensity category	"intensity_category" in d; d["intensity_category"] == s.intensity_category	Pass	0.001s
Legacy sequence_gauge_summary.json is deleted after split-mode write.	not legacy.exists()	Pass	0.040s
Corrupt file skipped	len(baselines) == 1; "GAUGE-OK" in baselines	Pass	0.002s

Test Description	Acceptance Criteria	Result	Time
Empty directory	<code>_load_gaugehd.baselines(d) == {}</code>	Pass	0.000s
Missing directory	<code>_load_gaugehd.baselines(tmp_path / "nonexistent") == {}</code>	Pass	0.000s
Missing mean level skipped	<code>len(baselines) == 0</code>	Pass	0.001s
Multiple gauges	<code>len(baselines) == 3</code>	Pass	0.003s
Valid gaugehd files	<code>"GAUGE-001" in baselines;</code> <code>baselines["GAUGE-001"]["mean_level"] == 1.25 (+1 more)</code>	Pass	0.002s
Sequence with malformed sequence_start doesn't crash.	<code>result["num_sequences"] == 5</code>	Pass	0.033s
Explicitly test malformed date by patching sequences post-generation.	<code>result["num_sequences"] == 3</code>	Pass	0.030s
No crash when gauge json absent	<code>isinstance(result, dict)</code>	Pass	0.002s
Num gauges zero without gauge json	<code>result["num_gauges"] == 0</code>	Pass	0.002s
Sequences still written without gauge json	<code>(tmp_path / SEQUENCES_FILENAME).exists()</code>	Pass	0.002s
gauge.json present but every record missing gauge_id → n_all == 0.	<code>isinstance(result, dict);</code> <code>result["num_gauges"] == 0 (+1 more)</code>	Pass	0.002s
Sequences still written when all gauges invalid	<code>(tmp_path / SEQUENCES_FILENAME).exists()</code>	Pass	0.002s
count=2001 with verbose=False prints ETA line at i=2000.	<code>result["num_sequences"] == 2001;</code> <code>"2,000" in out or "ETA" in out</code>	Pass	2.151s
Base level derived from alert	<code>r["base_level"] == pytest.approx(3.5 * 0.35)</code>	Pass	0.000s
Flood alert	<code>_parse_gauge(_G1)["flood_alert"] ==</code> <code>pytest.approx(3.5)</code>	Pass	0.000s
Flood warning	<code>_parse_gauge(_G1)["flood_warning"] ==</code> <code>pytest.approx(4.6)</code>	Pass	0.000s
Gauge id	<code>_parse_gauge(_G1)["gauge_id"] ==</code> <code>"GAUGE-test001"</code>	Pass	0.000s
Lat lon	<code>r["lat"] == pytest.approx(51.46);</code> <code>r["lon"] == pytest.approx(-0.30)</code>	Pass	0.000s
Missing alert returns none	<code>_parse_gauge(bad) is None</code>	Pass	0.000s
Missing gauge id returns none	<code>_parse_gauge(bad) is None</code>	Pass	0.000s
Missing lat returns none	<code>_parse_gauge(bad) is None</code>	Pass	0.000s
Severe warning	<code>_parse_gauge(_G1)["severe_warning"]</code> <code>== pytest.approx(5.5)</code>	Pass	0.000s
Valid record returns dict	<code>result is not None</code>	Pass	0.000s
Warning defaults when absent	<code>r is not None; r["flood_warning"] ==</code> <code>pytest.approx(4.0 * 1.33) (+1 more)</code>	Pass	0.000s

Test Description	Acceptance Criteria	Result	Time
Baseline without monthly means	<code>result["base_level"] == 1.5;</code>	Pass	0.000s
With baseline	<code>result["monthly_means"] is None</code> <code>result["base_level"] == 1.25;</code> <code>result["monthly_means"] == {"01": 1.4}</code>	Pass	0.000s
Without baseline uses heuristic	<code>result["base_level"] ==</code> <code>pytest.approx(3.5 * 0.35);</code> <code>result["monthly_means"] is None</code>	Pass	0.000s
Empty monthly means	<code>_seasonal_base_level({},</code> <code>mean_level=1.0, month=6) == 1.0</code>	Pass	0.000s
Missing month falls back	<code>_seasonal_base_level(monthly,</code> <code>mean_level=1.0, month=3) ==...</code>	Pass	0.000s
No monthly means	<code>_seasonal_base_level(None,</code> <code>mean_level=1.0, month=6) == 1.0</code>	Pass	0.000s
With monthly means	<code>_seasonal_base_level(monthly,</code> <code>mean_level=1.0, month=1) ==...;</code> <code>_seasonal_base_level(monthly,</code> <code>mean_level=1.0, month=7) ==...</code>	Pass	0.000s
All have intensity category	<code>all(r["intensity_category"] in valid</code> <code>for r in records)</code>	Pass	0.000s
All have sequence id	<code>all("sequence_id" in r for r in</code> <code>records)</code>	Pass	0.000s
All have sequence type	<code>all(r["sequence_type"] in valid for r</code> <code>in records)</code>	Pass	0.000s
All have total precip	<code>all("total_precip_mm" in r for r in</code> <code>records); all(r["total_precip_mm"] > 0</code> <code>for r in records)</code>	Pass	0.000s
Gauge file count matches num gauges	<code>len(gauge_records) == 3</code>	Pass	0.000s
Peak hours in valid range	<code>0 <= s["peak_hour"] <= 167</code>	Pass	0.000s
Peaks positive	<code>all(s["peak_m"] > 0 for s in seqs)</code>	Pass	0.000s
Per gauge records have alert flags	<code>all("alert" in s for s in seqs)</code>	Pass	0.000s
Per gauge records have peak	<code>all("peak_m" in s for s in seqs)</code>	Pass	0.000s
Per gauge records have peak hour	<code>all("peak_hour" in s for s in seqs)</code>	Pass	0.000s
Per gauge sequence count	<code>len(seqs) == len(records)</code>	Pass	0.000s
Severe implies warning	<code>s["warning"]</code>	Pass	0.000s
Every sequence that breached warning must also have breached alert.	<code>s["alert"]</code>	Pass	0.000s
Error message lists available ids	<code>"GAUGE-test001" in</code> <code>str(exc_info.value)</code>	Pass	0.001s
Invalid gauge id raises	—	Pass	0.002s

Test Description	Acceptance Criteria	Result	Time
Named file gauge ids	<code>d["gauge_ids"] == ["GAUGE-test002"]</code>	Pass	0.001s
Named file peaks length one	<code>all(len(r["peaks_m"]) == 1 for r in d["sequences"])</code>	Pass	0.001s
Named file schema version	<code>d["schema_version"] == SCHEMA_VERSION_SPATIAL</code>	Pass	0.004s
Named file sequence count	<code>len(d["sequences"]) == 40</code>	Pass	0.001s
Named output file written	<code>(gauge_dir / "sequence_gauge_GAUGE-test002.json").exists()</code>	Pass	0.000s
Summary num gauges is one	<code>single_run["num_gauges"] == 1</code>	Pass	0.000s
Single gauge classifier result dict	<code>isinstance(result, dict)</code>	Pass	6.234s
Single gauge + train_classifier=True covers lines 353-361.	<code>result["num_gauges"] == 1; stressm_dir.exists()</code>	Pass	6.301s
Base level matches gauge params	<code>responses["GAUGE-A"][0]["base_level_m"] == pytest.approx(...; responses["GAUGE-B"][0]["base_level_m"] == pytest.approx(...</code>	Pass	0.000s
Gauge a values	<code>r["base_level_m"] == pytest.approx(1.5); r["peak_level_m"] == pytest.approx(5.0) (+1 more)</code>	Pass	0.000s
Gauge b values	<code>r["base_level_m"] == pytest.approx(2.0); r["peak_level_m"] == pytest.approx(7.0) (+1 more)</code>	Pass	0.000s
Level change is peak minus base	<code>r["level_change_m"] == pytest.approx(expected)</code>	Pass	0.000s
Response has base level m	<code>"base_level_m" in r</code>	Pass	0.000s
Response has level change m	<code>"level_change_m" in r</code>	Pass	0.000s
Creates new summary	<code>summary["num_gauges"] == 1; summary["num_trained"] == 1 (+1 more)</code>	Pass	0.001s
Merges into existing	<code>summary["num_gauges"] == 2; summary["num_trained"] == 2 (+1 more)</code>	Pass	0.003s
Replaces existing gauge	<code>summary["num_gauges"] == 1; summary["gauges"][0]["metrics"]["auc_roc"] == 0.95</code>	Pass	0.002s
Skipped gauge not counted as trained	<code>summary["num_trained"] == 0; summary["num_skipped"] == 1</code>	Pass	0.001s
With count ; 2000 and verbose=False, no mid-loop progress line.	<code>"ETA" not in out or "[" not in out</code>	Pass	0.119s

Test Description	Acceptance Criteria	Result	Time
count=501 with verbose=True triggers the progress line at i=500.	<code>result["num_sequences"] == 501; "[500" in out or "500/" in out</code>	Pass	0.666s

1.1 Test Document History

Date	Version	Description	Author
27-Mar-2026	1.0	Initial test suite (126 tests)	D. Kelly
29-Mar-2026	1.1	19 test(s) removed (total: 107)	D. Kelly